In [2]:
```python
import pandas as pd
import numpy as np
```

importing dataset

In [13]:
```python
movies = pd.read_csv('bollywood.csv').squeeze()
```

In [10]:
```python
subs = pd.read_csv('subs.csv').squeeze()
```

In [11]:
```python
vk = pd.read_csv('kohli_ipl.csv').squeeze()
```

In [14]:
```python
movies.sample(2)
```

Out[14]:

	movie	lead
790	Aarakshan	Amitabh Bachchan
721	Chakravyuh (2012 film)	Arjun Rampal

In [15]:
```python
subs.sample(2)
```

Out[15]:
```
129    135
317    183
Name: Subscribers gained, dtype: int64
```

In [18]:
```python
vk.sample(2)
```

Out[18]:

	match_no	runs
14	15	50
172	173	29

In [27]:
```python
marks = {
    'maths':67,
    'english':57,
    'science':89,
    'hindi':100
}

marks_series = pd.Series(marks,name='marina ke marks')
marks_series
```

Out[27]:
```
maths       67
english     57
science     89
hindi      100
Name: marina ke marks, dtype: int64
```

In [22]:
```python
runs = [13,24,56,78,100]
pd.Series(runs)
```

```
Out[22]: 0     13
1     24
2     56
3     78
4    100
dtype: int64
```

EDITING SERIES

```python
In [28]: # editing using index
marks_series[0] = 100
marks_series
```

```
C:\Users\Admin\AppData\Local\Temp\ipykernel_11916\3909561698.py:2: FutureWarning: Series.__setitem__ treating keys as positions is deprecated. In a future version, integer keys will always be treated as labels (consistent with DataFrame behavior). To set a value by position, use `ser.iloc[pos] = value`
  marks_series[0] = 100
```

```
Out[28]: maths      100
english     57
science     89
hindi      100
Name: marina ke marks, dtype: int64
```

```python
In [29]: # waht if we don't  use wrong index??
marks_series['ece'] = 100
marks_series
```

```
Out[29]: maths      100
english     57
science     89
hindi      100
ece        100
Name: marina ke marks, dtype: int64
```

```python
In [31]: # editing using slicing
marks_series[2:4] = [90,90]
marks_series
```

```
Out[31]: maths      100
english     57
science     90
hindi       90
ece        100
Name: marina ke marks, dtype: int64
```

```python
In [36]: #using indexlabel
movies['Uri: The Surgical Strike'] = 'marina' ######## work on these ther
movies.sample()
```

Out[36]:

	movie	lead	Chakravyuh (2012 film)\t	Uri: The Surgical Strike
687	Arjun: The Warrior Prince	Yudhveer Bakoliya	ARJUN KAPOOR	marina

SERIES WITH PYTHON FUNCTIONALITIES

In [37]:
```
#LEN
print(len(subs))
```

```
365
```

In [38]:
```
#type
print(type(subs))
```

```
<class 'pandas.core.series.Series'>
```

In [40]:
```
#dir  ## dir shoews all the functions which we can perform to a dataset
print(dir(subs))
```

```
['T', '_AXIS_LEN', '_AXIS_ORDERS', '_AXIS_TO_AXIS_NUMBER', '_HANDLED_TYPE
S', '__abs__', '__add__', '__and__', '__annotations__', '__array__', '__ar
ray_priority__', '__array_ufunc__', '__bool__', '__class__', '__column_con
sortium_standard__', '__contains__', '__copy__', '__deepcopy__', '__delatt
r__', '__delitem__', '__dict__', '__dir__', '__divmod__', '__doc__', '__eq
__', '__finalize__', '__firstlineno__', '__float__', '__floordiv__', '__fo
rmat__', '__ge__', '__getattr__', '__getattribute__', '__getitem__', '__ge
tstate__', '__gt__', '__hash__', '__iadd__', '__iand__', '__ifloordiv__',
'__imod__', '__imul__', '__init__', '__init_subclass__', '__int__', '__inv
ert__', '__ior__', '__ipow__', '__isub__', '__iter__', '__itruediv__', '__
ixor__', '__le__', '__len__', '__lt__', '__matmul__', '__mod__', '__module
__', '__mul__', '__ne__', '__neg__', '__new__', '__nonzero__', '__or__',
'__pandas_priority__', '__pos__', '__pow__', '__radd__', '__rand__', '__rd
ivmod__', '__reduce__', '__reduce_ex__', '__repr__', '__rfloordiv__', '__r
matmul__', '__rmod__', '__rmul__', '__ror__', '__round__', '__rpow__', '__
rsub__', '__rtruediv__', '__rxor__', '__setattr__', '__setitem__', '__sets
tate__', '__sizeof__', '__static_attributes__', '__str__', '__sub__', '__s
ubclasshook__', '__truediv__', '__weakref__', '__xor__', '_accessors', '_a
ccum_func', '_agg_examples_doc', '_agg_see_also_doc', '_align_for_op', '_a
lign_frame', '_align_series', '_append', '_arith_method', '_as_manager',
'_attrs', '_binop', '_cacher', '_can_hold_na', '_check_inplace_and_allows_
duplicate_labels', '_check_is_chained_assignment_possible', '_check_label_
or_level_ambiguity', '_check_setitem_copy', '_clear_item_cache', '_clip_wi
th_one_bound', '_clip_with_scalar', '_cmp_method', '_consolidate', '_conso
lidate_inplace', '_construct_axes_dict', '_construct_result', '_constructo
r', '_constructor_expanddim', '_constructor_expanddim_from_mgr', '_constru
ctor_from_mgr', '_data', '_deprecate_downcast', '_dir_additions', '_dir_de
letions', '_drop_axis', '_drop_labels_or_levels', '_duplicated', '_find_va
lid_index', '_flags', '_flex_method', '_from_mgr', '_get_axis', '_get_axis
_name', '_get_axis_number', '_get_axis_resolvers', '_get_block_manager_axi
s', '_get_bool_data', '_get_cacher', '_get_cleaned_column_resolvers', '_ge
t_index_resolvers', '_get_label_or_level_values', '_get_numeric_data', '_g
et_rows_with_mask', '_get_value', '_get_values_tuple', '_get_with', '_geti
tem_slice', '_gotitem', '_hidden_attrs', '_indexed_same', '_info_axis', '_
info_axis_name', '_info_axis_number', '_init_dict', '_init_mgr', '_inplace
_method', '_internal_names', '_internal_names_set', '_is_cached', '_is_cop
y', '_is_label_or_level_reference', '_is_label_reference', '_is_level_refe
rence', '_is_mixed_type', '_is_view', '_is_view_after_cow_rules', '_item_c
ache', '_ixs', '_logical_func', '_logical_method', '_map_values', '_maybe_
update_cacher', '_memory_usage', '_metadata', '_mgr', '_min_count_stat_fun
ction', '_name', '_needs_reindex_multi', '_pad_or_backfill', '_protect_con
solidate', '_reduce', '_references', '_reindex_axes', '_reindex_indexer',
'_reindex_multi', '_reindex_with_indexers', '_rename', '_replace_single',
'_repr_data_resource_', '_repr_latex_', '_reset_cache', '_reset_cacher',
'_set_as_cached', '_set_axis', '_set_axis_name', '_set_axis_nocheck', '_se
t_is_copy', '_set_labels', '_set_name', '_set_value', '_set_values', '_set
_with', '_set_with_engine', '_shift_with_freq', '_slice', '_stat_functio
n', '_stat_function_ddof', '_take_with_is_copy', '_to_latex_via_styler',
'_typ', '_update_inplace', '_validate_dtype', '_values', '_where', 'abs',
'add', 'add_prefix', 'add_suffix', 'agg', 'aggregate', 'align', 'all', 'an
y', 'apply', 'argmax', 'argmin', 'argsort', 'array', 'asfreq', 'asof', 'as
type', 'at', 'at_time', 'attrs', 'autocorr', 'axes', 'backfill', 'betwee
n', 'between_time', 'bfill', 'bool', 'case_when', 'clip', 'combine', 'comb
ine_first', 'compare', 'convert_dtypes', 'copy', 'corr', 'count', 'cov',
'cummax', 'cummin', 'cumprod', 'cumsum', 'describe', 'diff', 'div', 'divid
e', 'divmod', 'dot', 'drop', 'drop_duplicates', 'droplevel', 'dropna', 'dt
ype', 'dtypes', 'duplicated', 'empty', 'eq', 'equals', 'ewm', 'expanding',
'explode', 'factorize', 'ffill', 'fillna', 'filter', 'first', 'first_valid
_index', 'flags', 'floordiv', 'ge', 'get', 'groupby', 'gt', 'hasnans', 'he
ad', 'hist', 'iat', 'idxmax', 'idxmin', 'iloc', 'index', 'infer_objects',
```

```
'info', 'interpolate', 'is_monotonic_decreasing', 'is_monotonic_increasin
g', 'is_unique', 'isin', 'isna', 'isnull', 'item', 'items', 'keys', 'kur
t', 'kurtosis', 'last', 'last_valid_index', 'le', 'list', 'loc', 'lt', 'ma
p', 'mask', 'max', 'mean', 'median', 'memory_usage', 'min', 'mod', 'mode',
'mul', 'multiply', 'name', 'nbytes', 'ndim', 'ne', 'nlargest', 'notna', 'n
otnull', 'nsmallest', 'nunique', 'pad', 'pct_change', 'pipe', 'plot', 'po
p', 'pow', 'prod', 'product', 'quantile', 'radd', 'rank', 'ravel', 'rdiv',
'rdivmod', 'reindex', 'reindex_like', 'rename', 'rename_axis', 'reorder_le
vels', 'repeat', 'replace', 'resample', 'reset_index', 'rfloordiv', 'rmo
d', 'rmul', 'rolling', 'round', 'rpow', 'rsub', 'rtruediv', 'sample', 'sea
rchsorted', 'sem', 'set_axis', 'set_flags', 'shape', 'shift', 'size', 'ske
w', 'sort_index', 'sort_values', 'squeeze', 'std', 'struct', 'sub', 'subtr
act', 'sum', 'swapaxes', 'swaplevel', 'tail', 'take', 'to_clipboard', 'to_
csv', 'to_dict', 'to_excel', 'to_frame', 'to_hdf', 'to_json', 'to_latex',
'to_list', 'to_markdown', 'to_numpy', 'to_period', 'to_pickle', 'to_sql',
'to_string', 'to_timestamp', 'to_xarray', 'transform', 'transpose', 'trued
iv', 'truncate', 'tz_convert', 'tz_localize', 'unique', 'unstack', 'updat
e', 'value_counts', 'values', 'var', 'view', 'where', 'xs']
```

In [42]:

```
## dir functionnnn

import math
dir(math)
```

```
Out[42]: ['__doc__',
 '__loader__',
 '__name__',
 '__package__',
 '__spec__',
 'acos',
 'acosh',
 'asin',
 'asinh',
 'atan',
 'atan2',
 'atanh',
 'cbrt',
 'ceil',
 'comb',
 'copysign',
 'cos',
 'cosh',
 'degrees',
 'dist',
 'e',
 'erf',
 'erfc',
 'exp',
 'exp2',
 'expm1',
 'fabs',
 'factorial',
 'floor',
 'fma',
 'fmod',
 'frexp',
 'fsum',
 'gamma',
 'gcd',
 'hypot',
 'inf',
 'isclose',
 'isfinite',
 'isinf',
 'isnan',
 'isqrt',
 'lcm',
 'ldexp',
 'lgamma',
 'log',
 'log10',
 'log1p',
 'log2',
 'modf',
 'nan',
 'nextafter',
 'perm',
 'pi',
 'pow',
 'prod',
 'radians',
 'remainder',
 'sin',
 'sinh',
```

```
'sqrt',
'sumprod',
'tan',
'tanh',
'tau',
'trunc',
'ulp']
```

In [46]:
```python
#sort
print(sorted(subs))
```

```
[33, 33, 35, 37, 39, 40, 40, 40, 40, 42, 42, 43, 44, 44, 44, 45, 46, 46, 4
8, 49, 49, 49, 49, 50, 50, 50, 51, 54, 56, 56, 56, 56, 57, 61, 62, 64, 65,
65, 66, 66, 66, 66, 67, 68, 70, 70, 70, 71, 71, 72, 72, 72, 72, 72, 73, 7
4, 74, 75, 76, 76, 76, 76, 77, 77, 78, 78, 78, 79, 79, 80, 80, 80, 81, 81,
82, 82, 83, 83, 83, 84, 84, 84, 85, 86, 86, 86, 87, 87, 87, 87, 88, 88, 8
8, 88, 88, 89, 89, 89, 90, 90, 90, 90, 91, 92, 92, 92, 93, 93, 93, 93, 95,
95, 96, 96, 96, 96, 97, 97, 98, 98, 99, 99, 100, 100, 100, 101, 101, 101,
102, 102, 103, 103, 104, 104, 104, 105, 105, 105, 105, 105, 105, 105, 105,
105, 108, 108, 108, 108, 108, 108, 109, 109, 110, 110, 110, 111, 111, 112,
113, 113, 113, 114, 114, 114, 114, 115, 115, 115, 115, 117, 117, 117, 118,
118, 119, 119, 119, 119, 120, 122, 123, 123, 123, 123, 123, 124, 125, 126,
127, 128, 128, 129, 130, 131, 131, 132, 132, 134, 134, 134, 135, 135, 136,
136, 136, 137, 138, 138, 138, 139, 140, 144, 145, 146, 146, 146, 146, 147,
149, 150, 150, 150, 150, 151, 152, 152, 152, 153, 153, 153, 154, 154, 154,
155, 155, 156, 156, 156, 156, 157, 157, 157, 157, 158, 158, 159, 159, 160,
160, 160, 160, 162, 164, 166, 167, 167, 168, 170, 170, 170, 170, 171, 172,
172, 173, 173, 173, 174, 174, 175, 175, 176, 176, 177, 178, 179, 179, 180,
180, 180, 182, 183, 183, 183, 184, 184, 184, 185, 185, 185, 185, 186, 186,
186, 188, 189, 190, 190, 192, 192, 192, 196, 196, 196, 197, 197, 202, 202,
202, 203, 204, 206, 207, 209, 210, 210, 211, 212, 213, 214, 216, 219, 220,
221, 221, 222, 222, 224, 225, 225, 226, 227, 228, 229, 230, 231, 233, 236,
236, 237, 241, 243, 244, 245, 247, 249, 254, 254, 258, 259, 259, 261, 261,
265, 267, 268, 269, 276, 276, 290, 295, 301, 306, 312, 396]
```

In [47]:
```python
#min
print(min(subs))
```

```
33
```

In [48]:
```python
print(max(subs))
```

```
396
```

In [49]:
```python
#type conversion
marks_series
```

Out[49]:
```
maths      100
english     57
science     90
hindi       90
ece        100
Name: marina ke marks, dtype: int64
```

In [50]:
```python
list(marks_series)
```

Out[50]:
```
[100, 57, 90, 90, 100]
```

In [51]:
```python
dict(marks_series)
```

Out[51]:
```
{'maths': np.int64(100),
 'english': np.int64(57),
 'science': np.int64(90),
 'hindi': np.int64(90),
 'ece': np.int64(100)}
```

In [55]:
```python
#loop
for i in marks_series.index :
    print(i)
```

```
maths
english
science
hindi
ece
```

In [56]:
```python
# Arithmetic Operators(Broadcasting)
100 + marks_series
```

Out[56]:
```
maths      200
english    157
science    190
hindi      190
ece        200
Name: marina ke marks, dtype: int64
```

In [58]:
```python
#relational operator
vk <20
```

Out[58]:

	match_no	runs
0	True	True
1	True	False
2	True	True
3	True	True
4	True	True
...	...	...
210	False	True
211	False	False
212	False	False
213	False	False
214	False	True

215 rows × 2 columns

BOOLEAN INDEXING ON SERIES

In [64]:
```python
# Find no of 50's and 100's scored by kohli
vk[ vk >= 50 ].size
```

Out[64]: 430

In [67]:
```
# find number of ducks
vk[vk == 0].size
```

Out[67]: 430

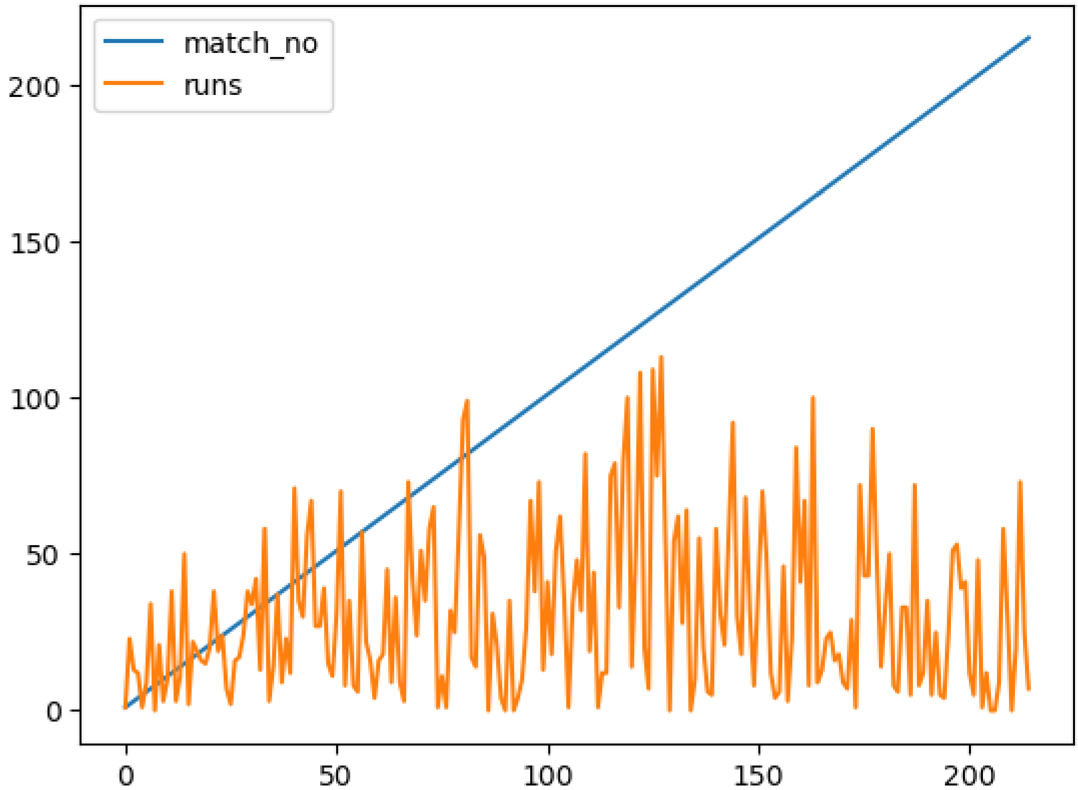
In [71]:
```
# Count number of day when I had more than 200 subs a day
subs[subs >=200].size
```

Out[71]: 59

PLOTTING GRAPH

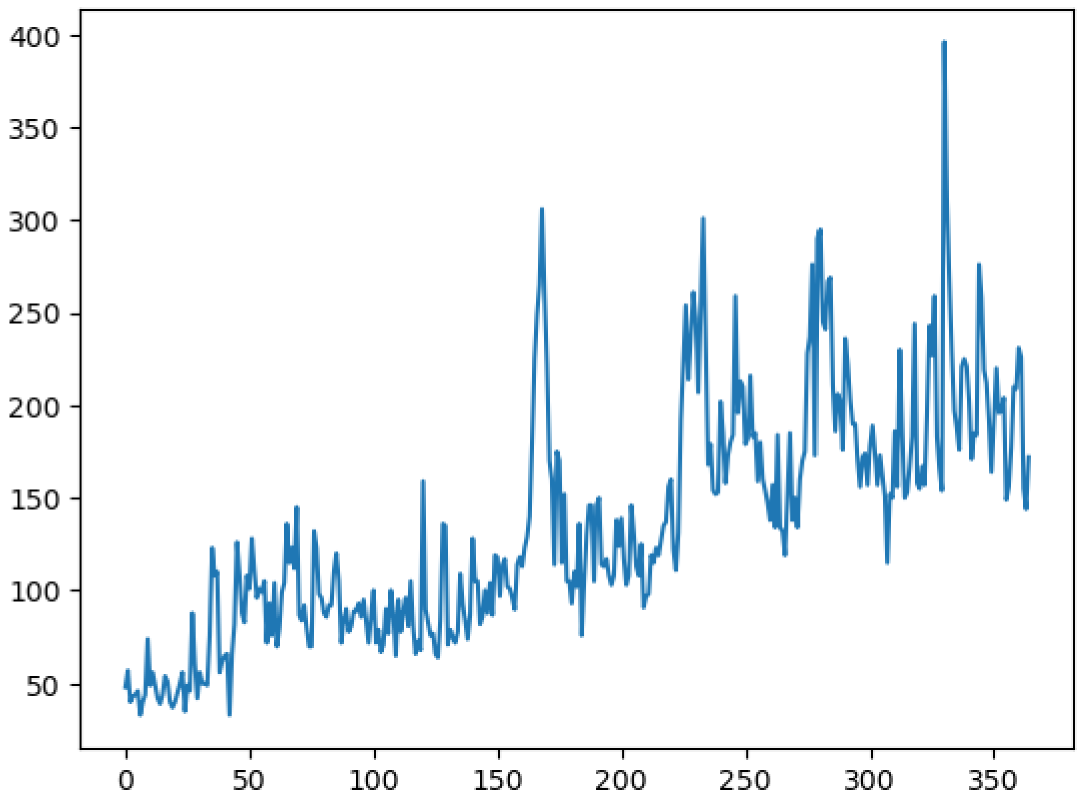
In [72]:
```
vk.plot()
```

Out[72]: <Axes: >



In [76]:
```
subs.plot()
```

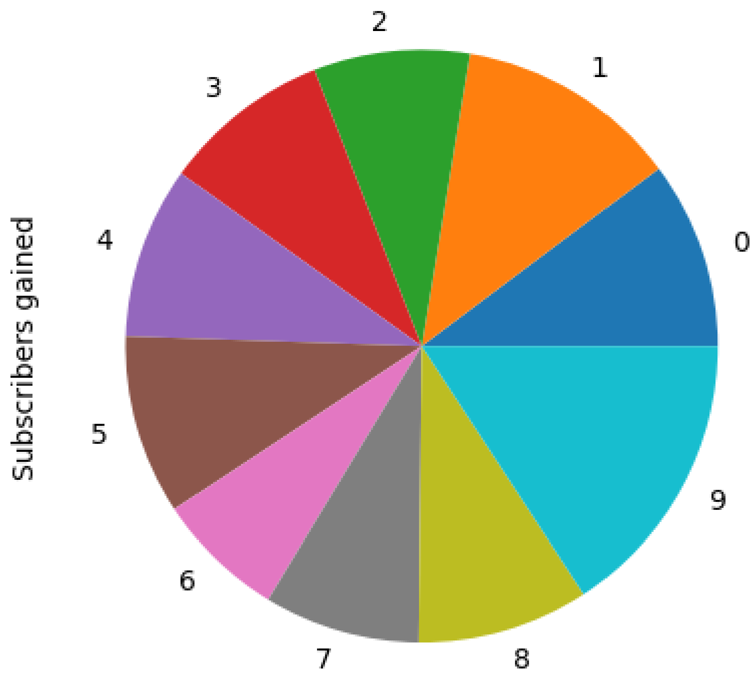
Out[76]: <Axes: >




```
In [78]: g1 = subs.head(10)
```

```
In [79]: g1.plot(kind = 'pie')
```

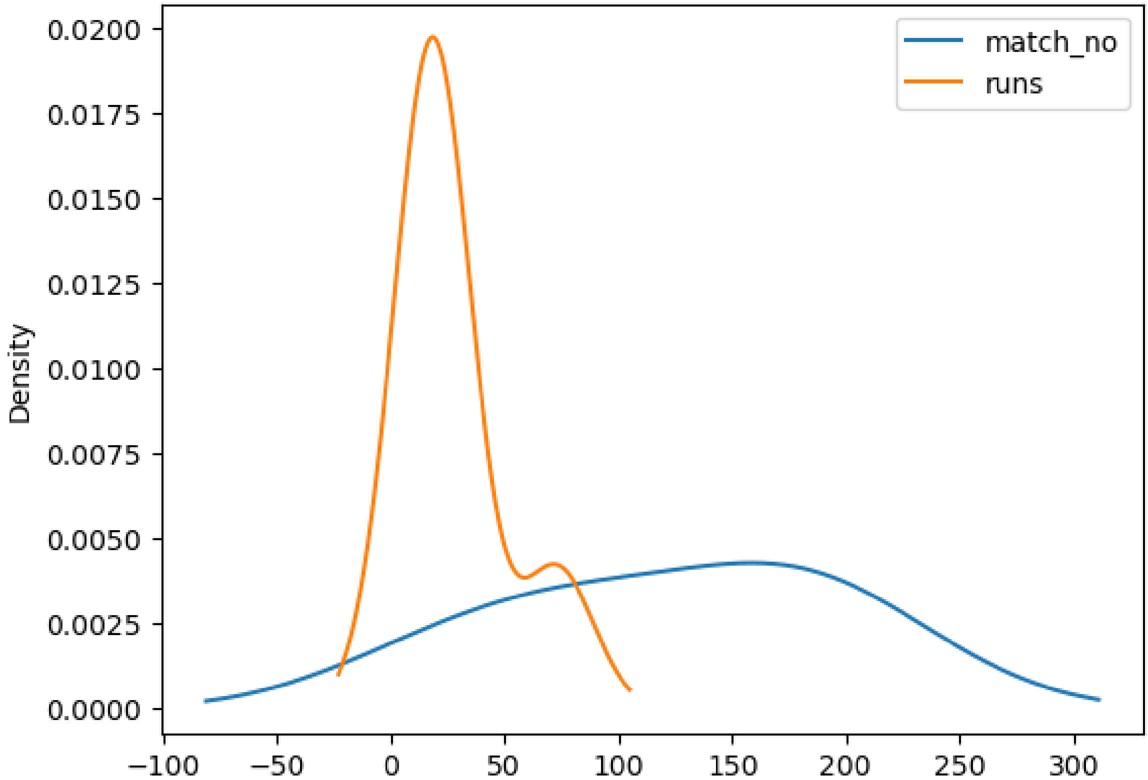
Out[79]: <Axes: ylabel='Subscribers gained'>




```
In [81]: g2 = vk.sample(6)
```

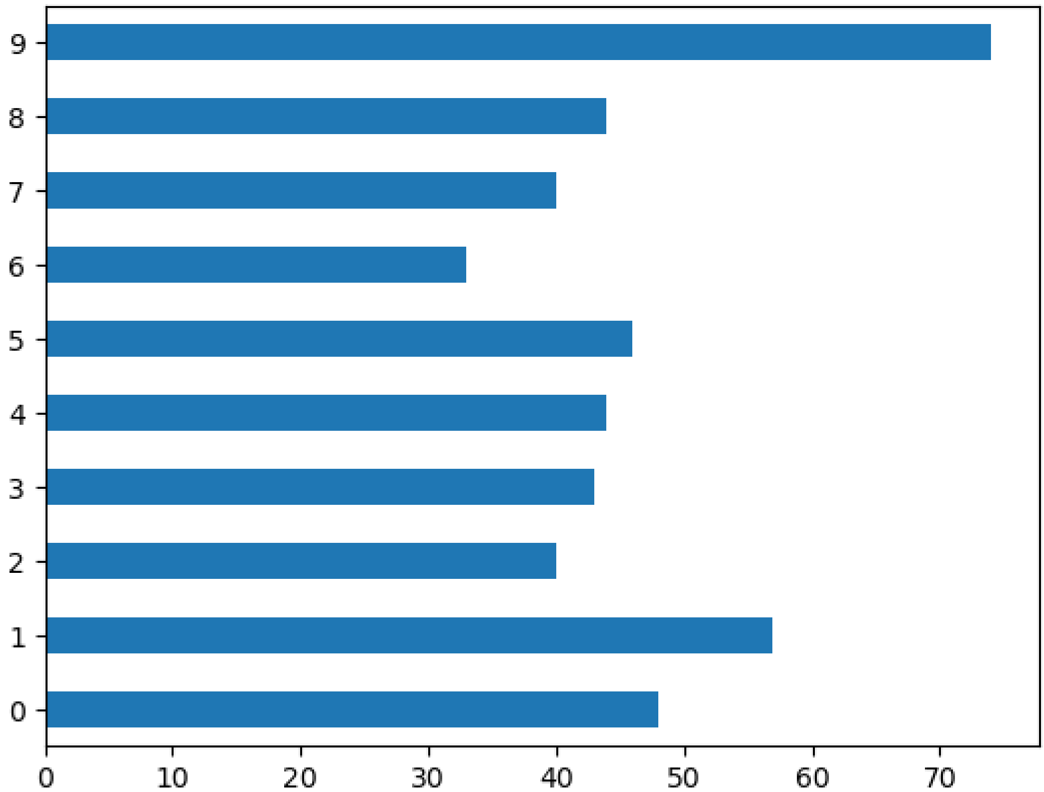
```
In [87]: g2.plot(kind = 'kde')
```

Out[87]: <Axes: ylabel='Density'>



In [88]:
```
g1.plot(kind = 'barh')
```

Out[88]: <Axes: >



In [89]:
```
g3 = vk.sample(9)
g3
```

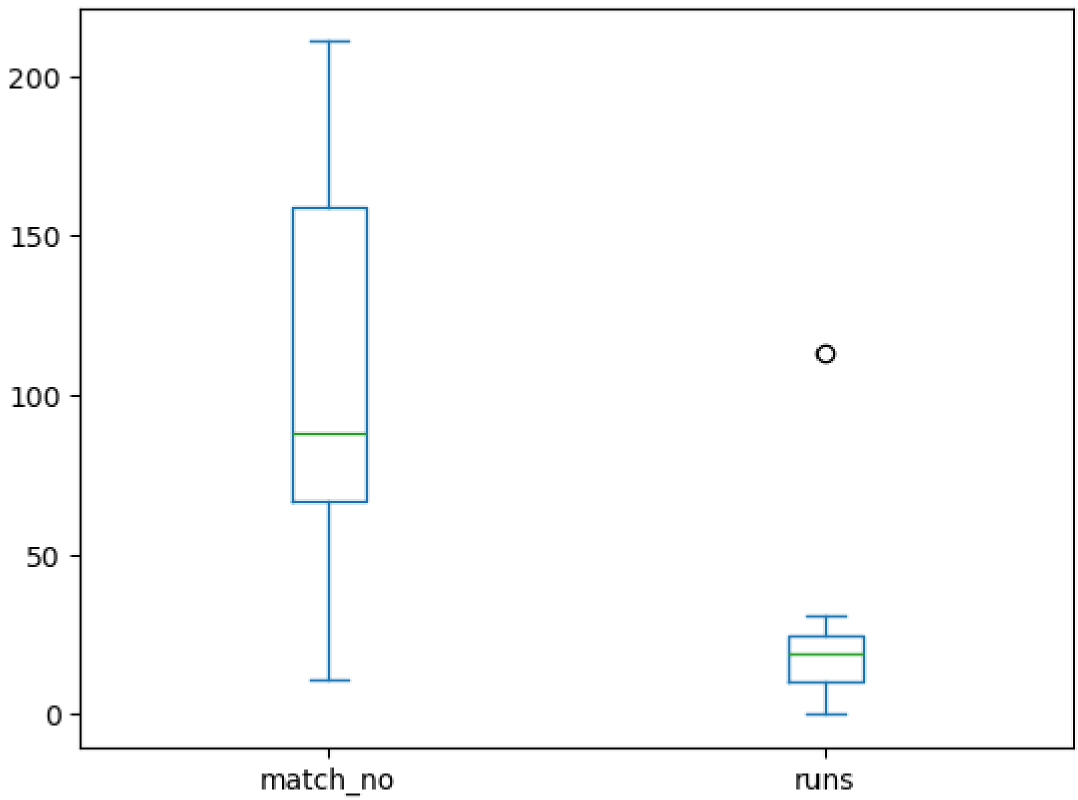
Out[89]:

	match_no	runs
87	88	31
78	79	25
10	11	10
158	159	23
22	23	19
127	128	113
210	211	0
66	67	3
165	166	13

In [91]:

```
g3.plot(kind = 's')
```

Out[91]: <Axes: >



In [ ]: